

Hugo Lebeau

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Family and environmental considerations influence my career choices.

Experience

Research

Inria, ENS Lyon (OCKHAM Team)

Postdoctoral Researcher

Theoretical study of compressive learning models.

Lyon, France

Feb. 2025–Present

Université Grenoble Alpes (LIG & GIPSA-lab)

Doctoral Researcher

Theoretical study of machine learning techniques with random matrix theory.

Grenoble, France

Oct. 2021–Jan. 2025

Université Grenoble Alpes (GIPSA-lab)

Research Intern

Analysis of online learning with random matrix theory.

Grenoble, France

Apr. 2021–Sep. 2021

CEA

Research Intern

Proximal algorithms to improve the quality of industrial neutron imaging.

Saclay, France

Mar. 2020–Jul. 2020

Politecnico di Milano

Research Intern

Statistical and numerical methods for functional data on complex multidimensional domains.

Milan, Italy

May 2019–Jun. 2019

Industry

AXA Climate

Data Scientist Intern

Weather data modeling and risk assessment for parametric insurance pricing.

Paris, France

Sep. 2019–Feb. 2020

Education

Université Grenoble Alpes (MIAI)

PhD in Applied Mathematics

“Teaching in Higher Education” track.

Grenoble, France

2021–2024

ENS Paris-Saclay

Master MVA (MSc) — Mathematics and Machine Learning

With honors of the jury.

Gif-sur-Yvette, France

2020–2021

ENSTA Paris

Diplôme d'Ingénieur (MSc) — Applied Mathematics, Optimization and Data Science

Ranked in the top 5% among 150 students.

Palaiseau, France

2017–2021

PhD Thesis: *Random Matrix and Tensor Models for Large Data Processing*

Manuscript: <https://theses.hal.science/tel-05222408v2>

Supervisors: Romain Couillet, Florent Chatelain.

Jury: Philippe Loubaton, Rémi Bardenet, Mylène Maïda, Walid Hachem, Olivier Michel, Pierre Comon.

Publications

Journals

- H. Lebeau, F. Chatelain, and R. Couillet, *A Random Matrix Approach to Low-Multilinear-Rank Tensor Approximation*, Journal of Machine Learning Research (JMLR), vol. 26, no. 7, pp. 1–64, 2025.

○ H. Lebeau, F. Chatelain, and R. Couillet, *Asymptotic Gaussian Fluctuations of Eigenvectors in Spectral Clustering*, IEEE Signal Processing Letters, vol. 31, pp. 1920–1924, 2024.

International Conferences.....
○ H. Lebeau, M. E. A. Seddik, and J. H. de M. Goulart, *Performance Gaps in Multi-view Clustering under the Nested Matrix-Tensor Model*, International Conference on Learning Representations (ICLR), 2024.
○ H. Lebeau, R. Couillet, and F. Chatelain, *A Random Matrix Analysis of Data Stream Clustering: Coping With Limited Memory Resources*, International Conference on Machine Learning (ICML), 2022.

National Conferences.....
○ H. Lebeau, *Performance of Rank-One Tensor Approximation on Incomplete Data*, Colloque GRETSI, 2025.
○ H. Lebeau, R. Couillet, and F. Chatelain, *HOSVD Tronquée : Analyse d'une Approximation Tensorielle Rapide*, Colloque GRETSI, 2023
○ H. Lebeau, R. Couillet, and F. Chatelain, *Une analyse par matrices aléatoires de l'apprentissage en ligne : traiter des grandes données avec des ressources mémoire limitées*, Colloque GRETSI, 2022.

Teaching

Random Matrix Theory and Machine Learning <i>Teaching Assistant, 9h/year</i> Introduction to the theory of large random matrices and their applications to machine learning with practical applications.	ENS Paris-Saclay, Master MVA <i>Since spring 2022</i>
Statistics <i>Teaching Assistant, 24h/year</i> Advanced statistics topics around parametric estimation, maximum likelihood estimation, testing, regression, concentration of random variables.	ENS Lyon, M1 <i>Spring 2025</i>
Probability and Statistics (STA401) <i>Teaching Assistant, 18h/year</i> Basics of probability, standard probability laws, descriptive statistics, estimation, hypothesis testing.	Université Grenoble Alpes, L2 <i>Spring 2023, 2024</i>
Introduction to Artificial Intelligence (INF103) <i>Teaching Assistant, 18h/year</i> Introduction to basic concepts of machine learning: datasets, classifiers, training, performance evaluation, data processing.	Université Grenoble Alpes, L1 <i>Fall 2021, 2022, 2023</i>
Introduction to Machine Learning <i>Teaching Assistant, 18h/year</i> Overview of the main tools in machine learning: model assessment, discriminant analysis, PCA, GLM and penalization, clustering with k -means and EM, trees and random forests, deep learning.	Grenoble INP, ENSE³, M2 <i>Fall 2022</i>
Functional Programming <i>Teaching Assistant, 36h/year</i> Introduction to functional programming with OCAML.	Université Grenoble Alpes, L1 <i>Spring 2022</i>

Technical skills

Programming languages
Proficient in: Python (Numpy, Scipy, Scikit-learn, Pytorch, Matplotlib)
Familiar with: R, Julia, MATLAB, C, C++, OCaml
Software
LaTeX, Git

Languages

French: Mother tongue	English: Professional	German: Intermediate
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Miscellaneous

○ *Sports & Endurance:* Road and trail running, cycling, and hiking. Deeply interested in sports science, specifically physiology, nutrition, and training methodologies.
○ *Performing Arts:* Theater enthusiast, practicing improvisational theater.
○ *Intellectual Interests:* Keen interest in physics, biology, philosophy, and French literature.